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2. Fold it once along its diameter to obtain two semicircles. Shade one half of the circle (Fig. 2.13)



Fig. 2.13

3. Again fold the semicircles to get 4 sectors. Fig. 2.14 shows a circle divided into four sectors. The sectors are re-arranged and made into a shape as shown in Fig. 2.15.

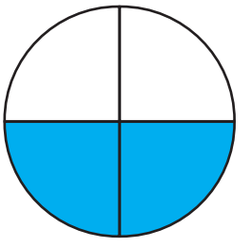


Fig. 2.14

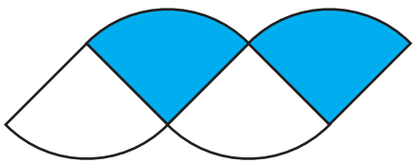


Fig. 2.15

4. Repeat this process of folding to eight folds, then it looks like a small sectors as shown in the Fig. 2.16. The sectors are re-arranged and made into a shape as shown in Fig. 2.17.

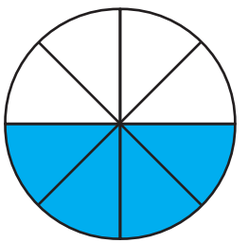


Fig. 2.16

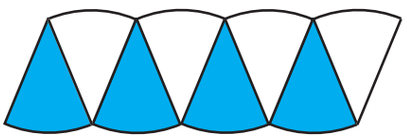


Fig. 2.17

5. Press and unfold the circle. It is then divided into 16 equal sectors and then into 32 equal sectors. As the number of sectors increase the assembled shape begins to look more or less like a rectangle, as shown in Fig. 2.19.

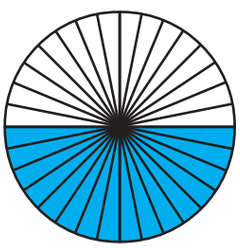


Fig. 2.18

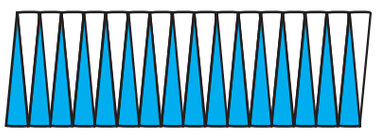


Fig. 2.19

6. The top and bottom of the rectangle is more or less equivalent to the circumference of the circle. Hence the top of the rectangle is half of the circumference $= \pi r$. The height of the rectangle is nearly equivalent to the radius of the circle. When the number of sectors is increased infinitely, the circle can be rearranged to form a rectangle of length 'πr' and breadth 'r'.

We know that the area of the rectangle $= l \times b$

$= \pi r \times r$

$= \pi r^2$

= Area of the circle

So, the area of the circle, $A = \pi r^2$ *sq.units* .

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